District-wise Analysis of Crimes Against Women in India (2014)

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Abstract

This paper presents a district-level analysis of crimes against women across India using the 2014 National Crime Records Bureau (NCRB) dataset, which contains 837 rows and 19 columns covering offence categories including rape, kidnapping, dowry deaths, domestic cruelty, and assault. After preprocessing to remove state-level summary rows, 801 genuine district records were retained. Exploratory data analysis reveals that Uttar Pradesh and West Bengal account for nearly 23% of the national total, and that cruelty by husband or relatives—captured under Section 498A IPC—constitutes approximately 40% of all recorded offences. A Linear Regression model trained on the four strongest predictors achieves an R^2 score of 0.965 on held-out data, demonstrating that a compact set of crime indicators can reliably estimate districtlevel crime burden.

Keywords: *crimes against women, NCRB, India, district-wise analysis, exploratory data analysis, linear regression, dowry deaths, domestic cruelty, kidnapping.*

I. Introduction

Each year, the National Crime Records Bureau (NCRB) publishes granular crime statistics disaggregated to the district level—a systematic record of where women face the most documented danger. The 2014 edition spans hundreds of districts and more than a dozen offence categories: rape, kidnapping, dowry deaths, domestic violence, assault, and others. Yet raw tabulations rarely surface the structural patterns within them. Which states bear a disproportionate share of recorded offences? Which crime types dominate the aggregate? And can a parsimonious predictive model reliably estimate overall districtlevel crime burden?

This paper addresses these questions through a two-stage methodology: exploratory data analysis (EDA) followed by supervised regression. The dataset is cleaned, visualised, and statistically characterised before a Linear Regression model is trained on the four strongest predictor variables. The objective is both analytical—understanding geographic and categorical patterns—and practical, since a reliable predictive indicator can serve as an early-warning mechanism for resource allocation.

An important methodological caveat frames the entire analysis: reported crime counts are an imperfect proxy for actual crime. High counts in some districts may partly reflect stronger reporting infrastructure or greater victim willingness to file formal complaints. Conversely, low counts in certain regions may signal institutional failure rather than genuine safety. These limitations are acknowledged throughout.

II. Dataset Description

The dataset originates from the NCRB, the nodal statistical agency under India's Ministry of Home Affairs, which compiles national crime data from state police departments. The 2014 release provides district-level records for all states and union territories. Table I summarises the dataset's principal attributes.

TABLE I. DATASET ATTRIBUTE SUMMARY

Attribute	Detail
Records	837 rows (incl. state summaries)
District Records (clean)	801 rows
Features	19 columns (2 categorical, 17 numeric)
Time Period	Year 2014
Source	NCRB, Govt. of India
Missing Values	None
Target Variable	Total Crimes against Women

The 17 numeric columns correspond to 13 distinct offence categories alongside derived aggregates. The principal crime types are: (1) Rape and Attempt to Commit Rape; (2) Kidnapping and Abduction; (3) Dowry Deaths; (4) Assault on Women with Intent to Outrage Modesty; (5) Insult to the Modesty of Women; (6) Cruelty by Husband or his Relatives; (7) Importation of Girls from Foreign Country; (8) Abetment of Suicides of Women; (9) Dowry Prohibition Act violations; (10) Indecent Representation of Women Act violations; (11) POCSO Act cases; (12) Protection of Women from Domestic Violence Act, 2005; and (13) Immoral Traffic Prevention Act cases.

III. Methodology

A. Data Preprocessing

The dataset was loaded using Pandas. An initial df.info() call confirmed complete data across all 837 rows—an unusual property that simplified preprocessing. The primary cleaning task was the removal of 'Total District(s)' summary rows that NCRB appends at the end of each state block; retaining these would have caused double-counting in all subsequent aggregations. Following their removal, 801 genuine district-level records remained. Serial number and year columns, carrying no analytical utility, were dropped.

B. Exploratory Data Analysis

Visualisations were constructed using Matplotlib and Seaborn prior to model development, as graphical inspection frequently surfaces patterns that formal statistics alone do not. The following charts were produced: (i) bar charts of the top 10 districts and top 10 states by total crimes; (ii) a histogram of the district-level crime distribution; (iii) a pie chart of the proportional contribution of the six most common crime types;

(iv) a line graph comparing trends across the five highest-burden states; (v) a pair plot examining inter-variable relationships among key crime categories; and (vi) a full correlation heatmap across all numeric features.

C. Feature Selection

The correlation heatmap provided the basis for feature selection. Four crime categories exhibited the highest Pearson correlations with the target variable (total crimes), as presented in Table II.

TABLE II. FEATURE CORRELATIONS WITH TARGET VARIABLE

Crime Category (Feature)	Correlation (r)
Cruelty by Husband or his Relatives	0.88
Assault on Women (Outrage of Modesty)	0.81
Kidnapping & Abduction (Total)	0.79
Rape	0.75

D. Machine Learning Model

Since the prediction target is a continuous numeric value, this is formulated as a regression problem. A standard Ordinary Least Squares Linear Regression model from scikit-learn was selected for its interpretability and as an appropriate baseline. The modelling pipeline proceeded as follows:

1) Feature Scaling: StandardScaler was applied to all four predictors to prevent high-magnitude variables from dominating the loss surface.
2) Train-Test Split: An 80/20 partition was applied with random_state=42 to ensure reproducibility.
3) Model Training: Linear Regression was fitted on the scaled training partition.
4) Evaluation: Model performance was assessed using Mean Absolute Error (MAE), Mean Squared Error (MSE), and the coefficient of determination (R^2).

IV. Results and Findings

A. Descriptive Statistics

Across all 801 genuine district records, a total of 3,37,922 crimes against women were documented in 2014. The arithmetic mean of 422 cases per district is, however, a misleading summary statistic: the standard deviation is 2,986, indicating extreme dispersion. The minimum recorded count is zero—almost certainly reflecting underreporting rather than actual absence of crime—while the maximum is 4,752 cases in South 24 Parganas, West Bengal. The distribution is severely right-skewed, with over 75% of districts reporting fewer than 580 crimes.

B. Geographic Distribution

State-level aggregation reveals pronounced geographic concentration. The ten highest-burden states are listed in Table III.

TABLE III. TOP 10 STATES BY TOTAL RECORDED CRIMES (2014)

Rank	State / UT	Total Crimes
1	Uttar Pradesh	38,467
2	West Bengal	38,299
3	Rajasthan	31,151
4	Madhya Pradesh	28,678
5	Maharashtra	26,693
6	Assam	19,139
7	Andhra Pradesh	16,512
8	Bihar	15,383
9	Delhi UT	15,265
10	Odisha	14,606

Uttar Pradesh and West Bengal together account for approximately 23% of the national total—a striking concentration given that India has 36 states and union territories. Population size is a partial but insufficient explanation: Tamil Nadu and Gujarat, with comparable or larger urban economies, appear considerably further down the list. At the district level, West Bengal dominates: South 24 Parganas (4,752), Murshidabad (4,078), and North 24 Parganas (3,175) rank in the top five nationally, alongside Mumbai Commissionerate (3,895) and Bengaluru City (3,077).

C. Crime Category Analysis

National-level aggregation by offence category (Table IV) reveals that domestic cruelty towers above all other crime types.

TABLE IV. NATIONAL CRIME COUNT BY CATEGORY (2014)

Crime Category	Count	Share (%)
Cruelty by Husband / Relatives	1,22,877	~40%
Assault (Outrage of Modesty)	82,235	~27%
Kidnapping & Abduction	57,311	~19%
Rape	36,735	~12%
Dowry Deaths	8,455	~2.8%
Attempt to Commit Rape	4,234	~1.4%

The dominance of cruelty by husband or relatives—over 1.22 lakh cases, constituting roughly 40% of all recorded offences—is the single most significant categorical finding. These cases are captured under Section 498A of the Indian Penal Code and encompass harassment, physical abuse, and emotional cruelty by a spouse or his family, frequently in conjunction with dowry demands. The implication is analytically important: the most prevalent documented crime against women in India in 2014 was occurring inside households, not in public spaces.

D. Model Performance

Linear Regression trained on the four selected features achieved the results shown in Table V.

TABLE V. MODEL EVALUATION METRICS (TEST SET)

Metric	Value	Interpretation
R^2 Score	0.9653	Excellent fit
MAE	46.72	Avg. error ~47 cases
MSE	8,953.18	Some large deviations

An R^2 of 0.965 indicates that just four predictor variables account for 96.5% of the variance in district-level total crime counts. This result is notable given the simplicity of the model. The MAE of approximately 47 cases is small relative to the observed range of district totals (0–4,752), confirming practical predictive utility. The MSE of 8,953 indicates some larger deviations at the high end of the distribution, consistent with the right-skewed nature of the data.

V. Discussion

The most consequential finding of this analysis is the categorical dominance of domestic cruelty. At 40% of all recorded offences, Section 498A cases represent not an exceptional phenomenon but a systemic one, embedded in household dynamics across large portions of the country. Because domestic abuse is notoriously underreported—victims face social stigma, economic dependence, and institutional barriers to filing complaints—the 1.22 lakh recorded cases almost certainly represent a fraction of actual incidence.

The geographic concentration in Uttar Pradesh, West Bengal, and Rajasthan prompts interpretive caution. Population size partially explains inter-state variation, but states such as Tamil Nadu and Karnataka have substantial populations yet appear considerably lower in the ranking. The residual variation is more plausibly attributable to a combination of socioeconomic factors (income levels, female literacy, employment patterns), prevailing gender norms, child sex ratios, and—critically—the capacity and willingness of local police and judicial institutions to register formal complaints. A low-ranking district might be safer, or it might simply have weaker reporting infrastructure.

The predictive success of the linear model carries a specific practical implication: if four crime categories can predict aggregate crime burden with 96.5% accuracy, then targeted data collection on these four indicators could provide an efficient early-warning signal for policymakers without requiring full administrative reporting cycles.

Several limitations constrain the study. The analysis covers a single year, precluding trend identification. No socioeconomic covariates—literacy rates, poverty headcount ratios, gender ratio—are incorporated, and their inclusion would likely improve both explanatory depth and model performance. The assumption that reported counts approximate actual incidence is a standard but consequential one, particularly for crimes with welldocumented underreporting.

VI. Conclusion

This paper has applied exploratory data analysis and machine learning to the 2014 NCRB district-level dataset on crimes against women in India. The principal findings are threefold. First, geographic concentration is pronounced: Uttar Pradesh, West Bengal, and Rajasthan account for a disproportionate share of national crime totals, with West Bengal dominating at the district level. Second, categorical composition reveals that domestic cruelty under Section 498A IPC constitutes approximately 40% of all recorded offences—the single largest category, and one that implicates intra-household dynamics rather than public-space crime. Third, a Linear Regression model trained on four highly correlated predictors achieves an R^2 of 0.965 on held-out test data, establishing that a parsimonious indicator set can reliably estimate overall district crime burden.

Future work should extend the analysis longitudinally across multiple NCRB reporting years to identify trend trajectories, incorporate district-level socioeconomic covariates to decompose drivers of variation, and evaluate ensemble regression methods that may better handle the distributional skewness observed in the data. Even in its current form, this baseline analysis offers a datadriven foundation for more precisely targeted policy responses to gender-based violence in India.

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